C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Subatomic Particles **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Relative mass of a proton				1
2	Relative charge of a proton				+1
3	Relative mass of a neutron				1
4	Relative charge of a neutron				0
5	Relative mass of an electron				1/1840 (≈ 0)
6	Relative charge of an electron				−1
7	Atomic number (*Z*)				Number of protons
8	Mass number (*A*)				Number of protons + neutrons
9	Isotopes				Same number of protons, different number of neutrons
10	Location of most of an atom's mass				Nucleus

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Relative Masses and Isotopes **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Relative atomic mass (A_r)				Weighted mean mass of an atom ÷ 1/12 mass of a ^{12}C atom
2	Relative isotopic mass				Mass of an atom of an isotope ÷ 1/12 mass of a ^{12}C atom
3	Relative molecular mass (M_r)				Mean mass of a molecule ÷ 1/12 mass of a ^{12}C atom
4	Standard used for relative masses				Carbon-12
5	Why isotopes have the same chemical properties				Same electron configuration
6	Property that differs between isotopes				Mass (density)
7	Type of mass spectrometer in the AQA spec				Time of flight (TOF)
8	Two ionisation methods in TOF				Electrospray and electron impact
9	Ion formed by electrospray ionisation of X				XH^+
10	Ion formed by electron impact ionisation of X				X^+

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: Mass Spectrometry and Orbitals **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Kinetic energy equation				KE = ½mv^2
2	Equation linking velocity, distance and time				$v = d \div t$
3	x-axis of a mass spectrum				m/z
4	y-axis of a mass spectrum				Relative abundance
5	Ions that reach the TOF detector first				Lightest (lowest m/z)
6	Charge of ions detected in TOF				Positive (1+)
7	Maximum electrons in an s subshell				2
8	Maximum electrons in a p subshell				6
9	Maximum electrons in a d subshell				10
10	Maximum electrons in one orbital				2 (opposite spins)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 4: Electron Configuration **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Shape of an s orbital				Spherical
2	Shape of a p orbital				Dumbbell
3	Number of orbitals in a p subshell				3
4	Number of orbitals in a d subshell				5
5	Maximum electrons in the third shell				18
6	Subshell filled first: 4s or 3d				4s
7	Subshell emptied first when ions form: 4s or 3d				4s
8	Two Period 4 exceptions to the filling order				Cr and Cu
9	Electron configuration of Cr				[Ar] $3d^5$ $4s^1$
10	Electron configuration of Cu				[Ar] $3d^{10}$ $4s^1$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 5: Ionisation Energy **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	First ionisation energy				Energy to remove 1 mol of electrons from 1 mol of gaseous atoms
2	Equation for the first ionisation energy of Na				$Na(g) \rightarrow Na^{+}(g) + e^{-}$
3	Second ionisation energy				Energy to remove 1 mol of electrons from 1 mol of gaseous 1+ ions
4	Units of ionisation energy				$kJ\ mol^{-1}$
5	Trend in first IE down a group				Decreases
6	General trend in first IE across a period				Increases
7	Period 3 element with lower first IE than Mg				Al
8	Why Al has a lower first IE than Mg				Outer electron in higher-energy 3p subshell
9	Period 3 element with lower first IE than P				S
10	Why S has a lower first IE than P				Paired 3p electrons repel

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 6: Factors, Blocks and Ions **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Three factors affecting ionisation energy				Nuclear charge, atomic radius, shielding
2	Evidence for electron shells				Large jumps in successive IEs
3	Evidence for subshells				Drops in first IE at Al and S
4	Block containing Ca				s
5	Block containing Fe				d
6	Block containing Br				p
7	Electron configuration of Fe^{2+}				[Ar] $3d^6$
8	Electron configuration of Fe^{3+}				[Ar] $3d^5$
9	Electron configuration of O^{2-}				$1s^2\ 2s^2\ 2p^6$
10	Isoelectronic				Same electron configuration

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.